Find the rotten ***tomato***

INTRO TO WIRESHARK

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WIRESHARK

SIGN-IN

"Security is a journey, not a destination."

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TODAY'S AGENDA

What is Wireshark?

Why cybersecurity uses it?

What is network traffic?

What Is a Packet?

Common Types of Protocol

Wireshark Basics

Wireshark Interface Tour

Interactive Lab Session

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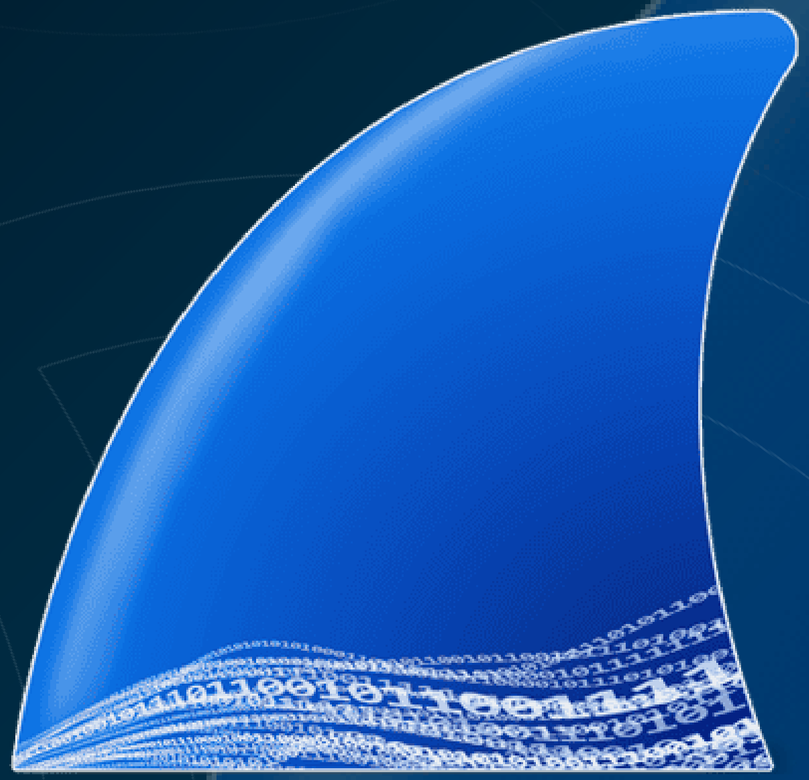
WHAT IS WIRESHARK?

- A network packet analyser, also known as a packet sniffer
- open-source
- lets you capture and inspect **network traffic**
- shows what is actually happening on a network

Think of it as an X-ray for the internet



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WHY PROFESSIONALS USE IT?

The Industry Standard for Network Insight

Who uses it?

SOC Analysts
Incident Responders
Malware Analysts
Network Security Engineers
Digital Forensics Investigators

Network Traffic Analysis (Blue Team)

Understand normal vs abnormal traffic
Validate IDS/SIEM alerts
Detect hidden threats

Incident Response

Investigate suspicious activity
Trace attacker IP

Malware Analysis & Reverse Engineering

Where malware connects
What data it sends
How often it communicates
Identify weird user-agent strings

Validating Security Controls

Confirm encryption (TLS/VPN)
Test firewall rules
Ensure policies are enforced

Phishing & Malware Detection

Suspicious domains
Strange URLs
Repeated traffic patterns

Digital Forensics & Evidence

Breach investigations
Timeline reconstruction

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WHAT IS NETWORK TRAFFIC?

- the flow of data between devices over a network

Any time you.....
Open a website
Send an email
Watch a video
Log in to an app
.....data is moving across the network.

So what Is actually **moving**? ..... **packets**

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WHAT IS PACKET?

packet = envelope

A packet is a small unit of data sent across a network.

Each packet contains:

- Source (where it came from)
- Destination (where it's going)
- Protocol (rules being used)
- Payload (the actual data)

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COMMON TYPES OF NETWORK PROTOCOL

HTTP / HTTPS – Web browsing

DNS – Translating website names to IP addresses

SMTP – Sending emails

TCP / UDP – How data is delivered

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WIRESHARK BASICS

Used for analysis, not hacking

How Wireshark Works (High Level)

Network traffic is captured → Traffic is broken into packets → Wireshark displays each packet → You analyze patterns and details

It does not modify traffic, it only observes.

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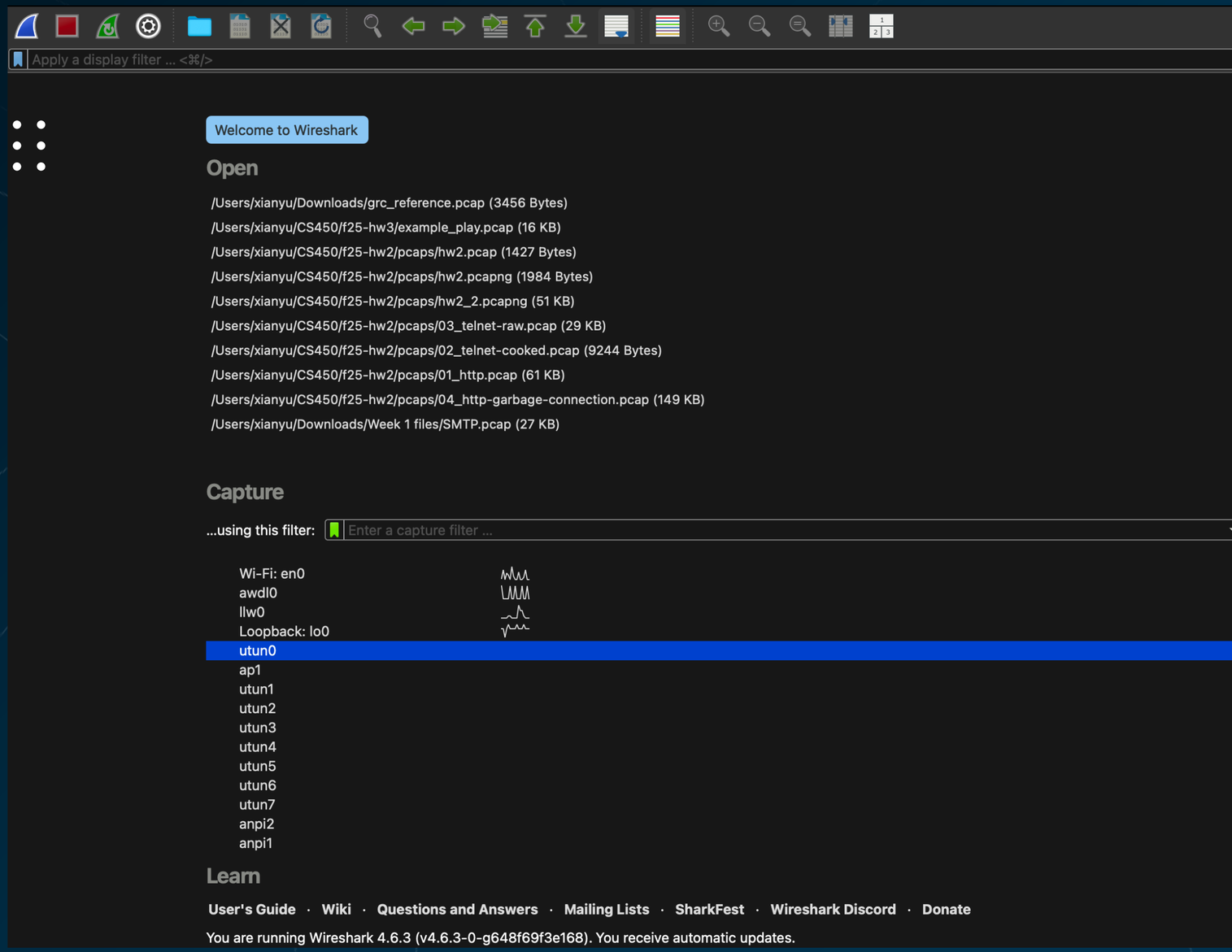
STARTING WIRESHARK

When you open Wireshark, you can:

- Choose a network interface (Wi-Fi, Ethernet)
- Open a PCAP file
- Apply display filters



Welcome to Wireshark

Open

/Users/xianyu/Downloads/grc_reference.pcap (3456 Bytes)
/Users/xianyu/CS450/f25-hw3/example_play.pcap (16 KB)
/Users/xianyu/CS450/f25-hw2/pcaps/hw2.pcap (1427 Bytes)
/Users/xianyu/CS450/f25-hw2/pcaps/hw2.pcapng (1984 Bytes)
/Users/xianyu/CS450/f25-hw2/pcaps/hw2_2.pcapng (51 KB)
/Users/xianyu/CS450/f25-hw2/pcaps/03_telnet-raw.pcap (29 KB)
/Users/xianyu/CS450/f25-hw2/pcaps/02_telnet-cooked.pcap (9244 Bytes)
/Users/xianyu/CS450/f25-hw2/pcaps/01_http.pcap (61 KB)
/Users/xianyu/CS450/f25-hw2/pcaps/04_http-garbage-connection.pcap (149 KB)
/Users/xianyu/Downloads/Week 1 files/SMTP.pcap (27 KB)

Capture

...using this filter: Enter a capture filter ...

Wi-Fi: en0
awdl0
llw0
Loopback: lo0
utun0
ap1
utun1
utun2
utun3
utun4
utun5
utun6
utun7
anpi2
anpi1

Learn

User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate

You are running Wireshark 4.6.3 (v4.6.3-0-g648f69f3e168). You receive automatic updates.

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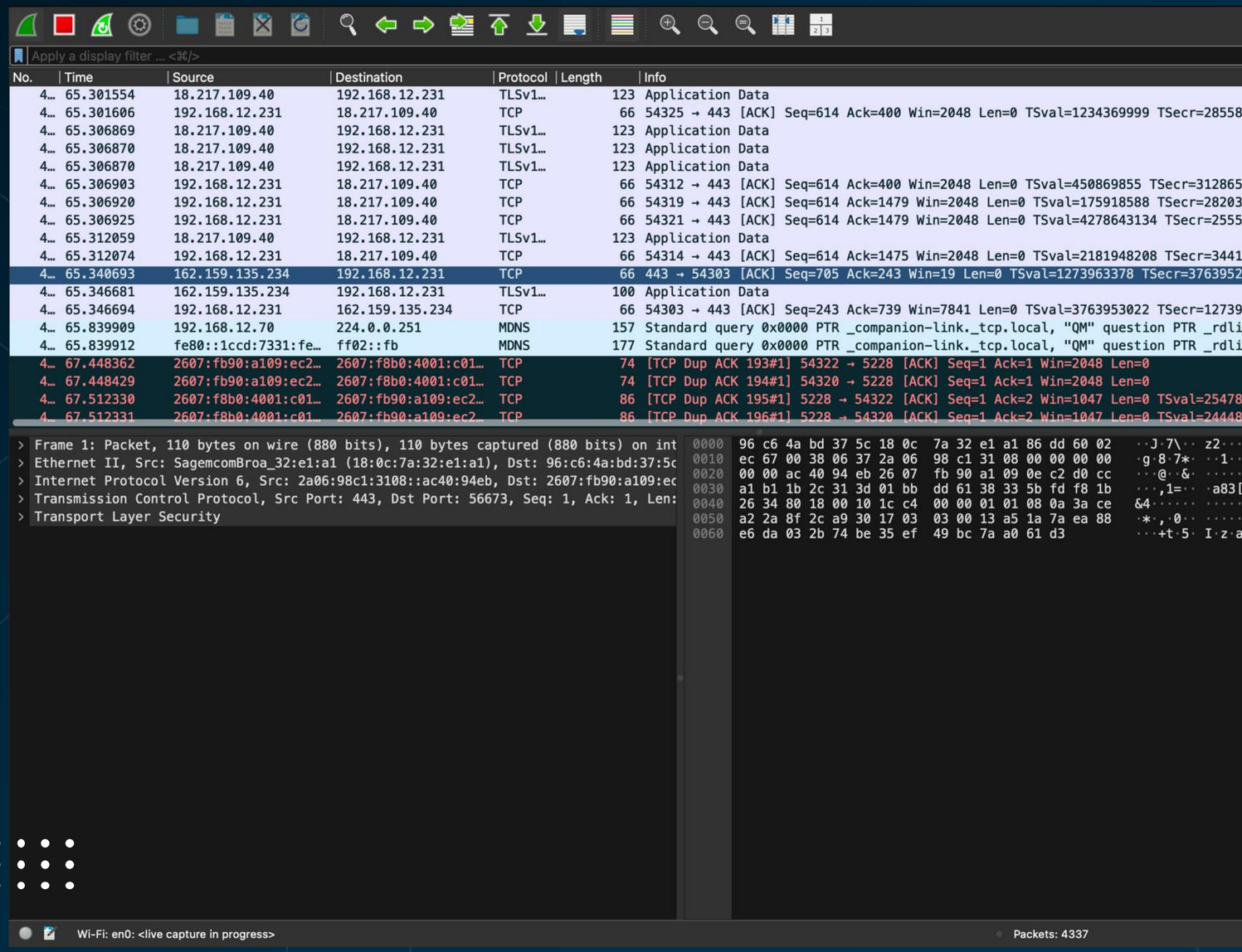
WIRESHARK INTERFACE TOUR

Wireshark has three main panes:

- Packet List Pane (Top)
- Packet Details Pane (Bottom Left)
- Packet Bytes Pane (Bottom Right)

Power Features:

- Display filters (find what you want)
- Follow TCP Stream (rebuild conversations)
- Color coding (spot patterns quickly)



No.	Time	Source	Destination	Protocol	Length	Info
4…	65.301554	18.217.109.40	192.168.12.231	TLSv1…	123	Application Data
4…	65.301606	192.168.12.231	18.217.109.40	TCP	66	54325 → 443 [ACK] Seq=614 Ack=400 Win=2048 Len=0 TSval=1234369999 TSecr=28558
4…	65.306869	18.217.109.40	192.168.12.231	TLSv1…	123	Application Data
4…	65.306870	18.217.109.40	192.168.12.231	TLSv1…	123	Application Data
4…	65.306870	18.217.109.40	192.168.12.231	TLSv1…	123	Application Data
4…	65.306903	192.168.12.231	18.217.109.40	TCP	66	54312 → 443 [ACK] Seq=614 Ack=400 Win=2048 Len=0 TSval=450869855 TSecr=312865
4…	65.306920	192.168.12.231	18.217.109.40	TCP	66	54319 → 443 [ACK] Seq=614 Ack=1479 Win=2048 Len=0 TSval=175918588 TSecr=28203
4…	65.306925	192.168.12.231	18.217.109.40	TCP	66	54321 → 443 [ACK] Seq=614 Ack=1479 Win=2048 Len=0 TSval=4278643134 TSecr=2555
4…	65.312059	18.217.109.40	192.168.12.231	TLSv1…	123	Application Data
4…	65.312074	192.168.12.231	18.217.109.40	TCP	66	54314 → 443 [ACK] Seq=614 Ack=1475 Win=2048 Len=0 TSval=2181948208 TSecr=3441
4…	65.340693	162.159.135.234	192.168.12.231	TCP	66	443 → 54303 [ACK] Seq=705 Ack=243 Win=19 Len=0 TSval=1273963378 TSecr=3763952
4…	65.346681	162.159.135.234	192.168.12.231	TLSv1…	100	Application Data
4…	65.346694	192.168.12.231	162.159.135.234	TCP	66	54303 → 443 [ACK] Seq=243 Ack=739 Win=7841 Len=0 TSval=3763953022 TSecr=12739
4…	65.839909	192.168.12.70	224.0.0.251	MDNS	157	Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR _rdli
4…	65.839912	fe80::1ccd:7331:fe…	ff02::fb	MDNS	177	Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR _rdli
4…	67.448362	2607:fb90:a109:ec2…	2607:f8b0:4001:c01…	TCP	74	[TCP Dup ACK 193#1] 54322 → 5228 [ACK] Seq=1 Ack=1 Win=2048 Len=0
4…	67.448429	2607:fb90:a109:ec2…	2607:f8b0:4001:c01…	TCP	74	[TCP Dup ACK 194#1] 54320 → 5228 [ACK] Seq=1 Ack=1 Win=2048 Len=0
4…	67.512330	2607:f8b0:4001:c01…	2607:fb90:a109:ec2…	TCP	86	[TCP Dup ACK 195#1] 5228 → 54322 [ACK] Seq=1 Ack=2 Win=1047 Len=0 TSval=25478
4…	67.512331	2607:f8b0:4001:c01…	2607:fb90:a109:ec2…	TCP	86	[TCP Dup ACK 196#1] 5228 → 54320 [ACK] Seq=1 Ack=2 Win=1047 Len=0 TSval=24448

```
> Frame 1: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on int
> Ethernet II, Src: SagemcomBroa_32:e1:a1 (18:0c:7a:32:e1:a1), Dst: 96:c6:4a:bd:37:5c
> Internet Protocol Version 6, Src: 2a06:98c1:3108::ac40:94eb, Dst: 2607:fb90:a109:ec
> Transmission Control Protocol, Src Port: 443, Dst Port: 56673, Seq: 1, Ack: 1, Len:
> Transport Layer Security
```

```
0000  96 c6 4a bd 37 5c 18 0c  7a 32 e1 a1 86 dd 60 02   ··J·7\·· z2····
0010  ec 67 00 38 06 37 2a 06  98 c1 31 08 00 00 00 00   ·g·8·7*· ··1···
0020  00 00 ac 40 94 eb 26 07  fb 90 a1 09 0e c2 d0 cc   ···@··&· ····
0030  a1 b1 1b 2c 31 3d 01 bb  dd 61 38 33 5b fd f8 1b   ···,1=·· ·a83[
0040  26 34 80 18 00 10 1c c4  00 00 01 01 08 0a 3a ce   &4····· ····
0050  a2 2a 8f 2c a9 30 17 03  03 00 13 a5 1a 7a ea 88   ·*·,·0·· ····
0060  e6 da 03 2b 74 be 35 ef  49 bc 7a a0 61 d3         ···+t·5· I·z·a
```

Wi-Fi: en0: <live capture in progress>

Packets: 4337

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LAB

Question 1: Can you find the bad tomato?

Question 2: How many people are attacked?

CheatSheet

https://tinyurl.com/kk5jy34u

Lab pcap file

https://tinyurl.com/mrx36nw3

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SOME USEFUL LINKS:

Download Wireshark: https://www.wireshark.org/download.html
User Guide: https://www.wireshark.org/docs/wsug_html_chunked/
SMTP Reference: https://www.wireshark.org/docs/dfref/s/smtp.html
Extra exercise website: https://www.malware-traffic-analysis.net/
tshark (terminal-based wireshark):
https://www.wireshark.org/docs/wsug_html_chunked/AppToolstshark.html

Pro Tips for Beginners:

You don't need to understand every packet
Start with filters, not raw traffic
Focus on patterns, not single packets
Ask: Who talked to whom, and why?

Signs of Suspicious Traffic:

Misspelled or strange domains
IP addresses instead of domain names
HTTP login pages (not HTTPS)
Repeated traffic patterns
Plaintext usernames or passwords

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THANK YOU FOR YOUR ATTENTION

Hope to see you again in our next event!

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